

# Pharmacodynamic features of Fluoxetine

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AT2 Oral Presentation

HMM301, 06/05/2026

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# Introduction



## Drug Identity

Name – Fluoxetine  
Brand name – Prozac

- Fluoxetine is a clinically used antidepressant.
- It belongs to the SSRI class and is given orally for depression and anxiety-related disorders.

## Disease treated

Fluoxetine is used to treat MDD (Major Depressive Disorder), a condition involving persistent low mood, loss of interest, reduced motivation, sleep/appetite changes, and impaired daily functioning.

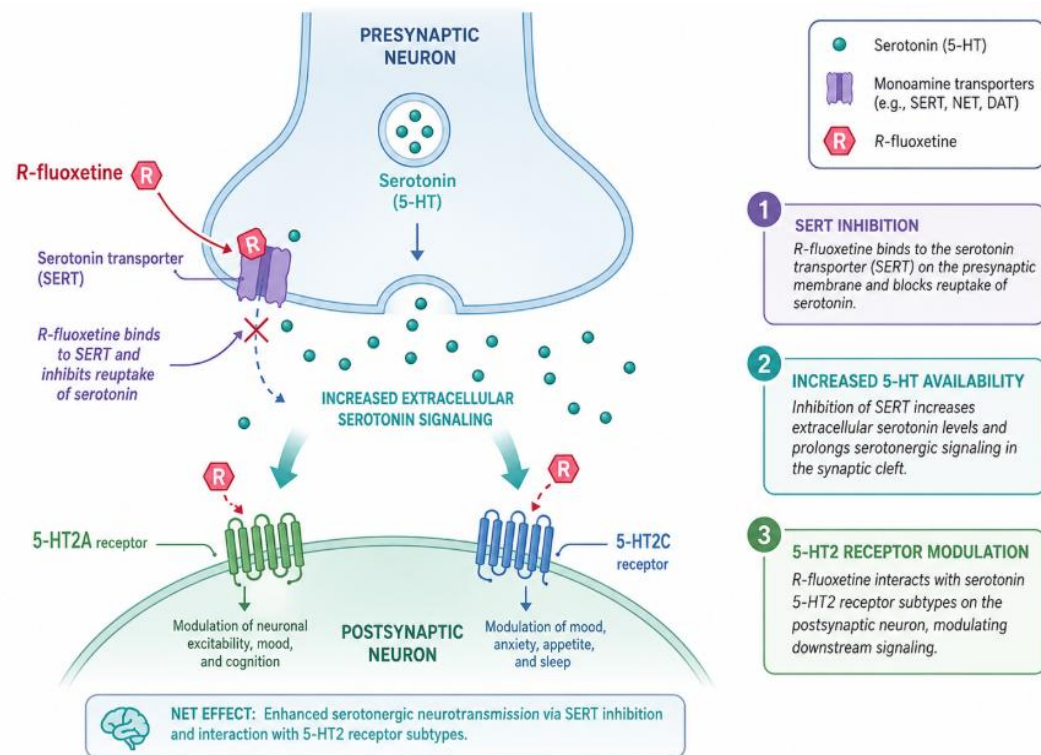
## Drug class

*Selective serotonin reuptake inhibitor (SSRI)*

Fluoxetine mainly targets serotonin signalling. This makes it more selective than older antidepressants such as tricyclic antidepressants (TCAs), which also interact with adrenergic, cholinergic, and histaminergic systems and can cause more side effects.

## Main pharmacodynamic action

Fluoxetine blocks the serotonin transporter, SERT, which normally takes serotonin back into the presynaptic nerve terminal. By inhibiting this reuptake process, fluoxetine allows more serotonin to remain in the extracellular space, enhancing serotonergic neurotransmission.



AI-generated schematic of R-fluoxetine's cellular/molecular mechanism. Created using ChatGPT image generation and adapted from mechanisms reported by Koch, S. (2002).

# Article I: Mechanisms of drug action at the cellular level

Fig 1: Transporter binding profile of fluoxetine and norfluoxetine enantiomers

| Compound          | Transporter |                     |           |
|-------------------|-------------|---------------------|-----------|
|                   | 5-HT        | NE                  | DA        |
|                   |             | K <sub>i</sub> , nM |           |
| R-fluoxetine      | 7.7 ± 1.0   | >10000              | >10000    |
| R-norfluoxetine   | 68 ± 12     | >10000              | >10000    |
| S-fluoxetine      | 6.1 ± 0.3   | >10000              | >10000    |
| S-norfluoxetine   | 4.8 ± 0.3   | >10000              | >10000    |
| R,S-fluoxetine    | 6.9 ± 0.6   | 1021 ± 46           | 4752 ± 44 |
| R,S-norfluoxetine | 9.9 ± 0.5   | >10000              | >10000    |

Modified from (Malagié et al., 1995), Table 1



## Method : Transporter binding assay

- Researchers used cell membranes engineered to express human monoamine transporters:

| Radioligand                  | Binds to                         |
|------------------------------|----------------------------------|
| [ <sup>3</sup> H]-paroxetine | serotonin transporter / SERT     |
| [ <sup>3</sup> H]-nisoxetine | norepinephrine transporter / NET |
| [ <sup>3</sup> H]-Win35,428  | dopamine transporter / DAT       |

- Then they added increasing concentrations of Fluoxetine.
- If fluoxetine binds strongly to SERT, it will compete with the radiolabeled paroxetine and push it off the transporter. The same principle applies to the other transporters.
- They measured how much radioactive ligand was still bound.

Equation: 
$$K_i = \frac{IC_{50}}{1 + \frac{[L]}{K_d}}$$

K<sub>i</sub> = binding affinity of fluoxetine for the transporter  
IC<sub>50</sub> = concentration of Fluoxetine needed to reduce radioligand binding by 50%  
[L] = concentration of the radiolabeled ligand  
K<sub>d</sub> = binding affinity of the radiolabeled ligand

Low K<sub>i</sub> means strong binding

High K<sub>i</sub> means weak binding

### Conclusion:

Fluoxetine and norfluoxetine bind much more strongly to the serotonin transporter than to norepinephrine or dopamine transporters, confirming that fluoxetine acts primarily as a SSRI.

# Article 1: Mechanisms of drug action at the cellular level

Fig 2: Agonist and Antagonist Activity of R-fluoxetine at Human 5HT<sub>2</sub> Cloned Receptors using GTP $\gamma$ S Binding

| Receptor           | % Max of Full Agonist | K <sub>i</sub> , nM |
|--------------------|-----------------------|---------------------|
| 5-HT <sub>2A</sub> | -10 $\pm$ 4           | 193 (107-346)       |
| 5-HT <sub>2B</sub> | 2 $\pm$ 6             | >10000              |
| 5-HT <sub>2C</sub> | -1 $\pm$ 5            | 493 (255-953)       |



**Result 1:** R-fluoxetine did not act as an agonist

**Result 2:**  
R-fluoxetine blocked:  
•5-HT<sub>2A</sub> receptor signaling  
•5-HT<sub>2C</sub> receptor signaling  
But it did **not** meaningfully block 5-HT<sub>2B</sub>

## Method: Functional 5-HT<sub>2</sub> receptor assay

- Human 5-HT<sub>2A</sub>, 5-HT<sub>2B</sub> and 5-HT<sub>2C</sub> receptors were expressed in cloned laboratory cell systems.
- These receptors are G-protein-coupled receptors, meaning receptor activation triggers intracellular G-protein signalling.
- The researchers used radioactive **GTP $\gamma$ [<sup>35</sup>S]** binding assay.
- When a receptor is activated, the associated G-protein binds GTP $\gamma$ S, so increased GTP $\gamma$ S binding indicates receptor activation.
- R-fluoxetine was first tested alone to determine whether it acted as an agonist. This was reported as % max of full agonist.
- Receptors were then activated with serotonin/5-HT, and R-fluoxetine was tested for its ability to block serotonin-stimulated signalling (antagonist effect)

### How K<sub>i</sub> was used

- The blocking strength of R-fluoxetine was reported as K<sub>i</sub>.
- In this table, K<sub>i</sub> represents how strongly R-fluoxetine blocks serotonin-stimulated 5-HT<sub>2</sub> receptor signaling.
- Lower K<sub>i</sub> = stronger antagonism.
- Higher K<sub>i</sub> = weaker antagonism.

# Article 2: Mechanisms of drug action at the tissues, organ or physiological level

Method: Measuring brain serotonin using in vivo microdialysis

## Experimental model

- Vivo model - Sprague-Dawley rats
- Aim - Measure how acute fluoxetine changes extracellular serotonin/5-HT levels in the brain.
- Rats were anaesthetized and placed in a stereotaxic frame to allow accurate probe placement.



## Brain regions tested

Microdialysis probes were implanted into three serotonergic brain regions:

**Frontal cortex**  
**Ventral hippocampus**  
**Raphe nuclei**

## Microdialysis procedure

- A microdialysis probe acts like a tiny sampling tube placed into brain tissue.
- Artificial cerebrospinal fluid (CSF) was continuously passed through the probe.
- Extracellular serotonin from the surrounding brain tissue diffused into the probe and was collected as dialysate.
- Dialysate samples were collected every 15 minutes.

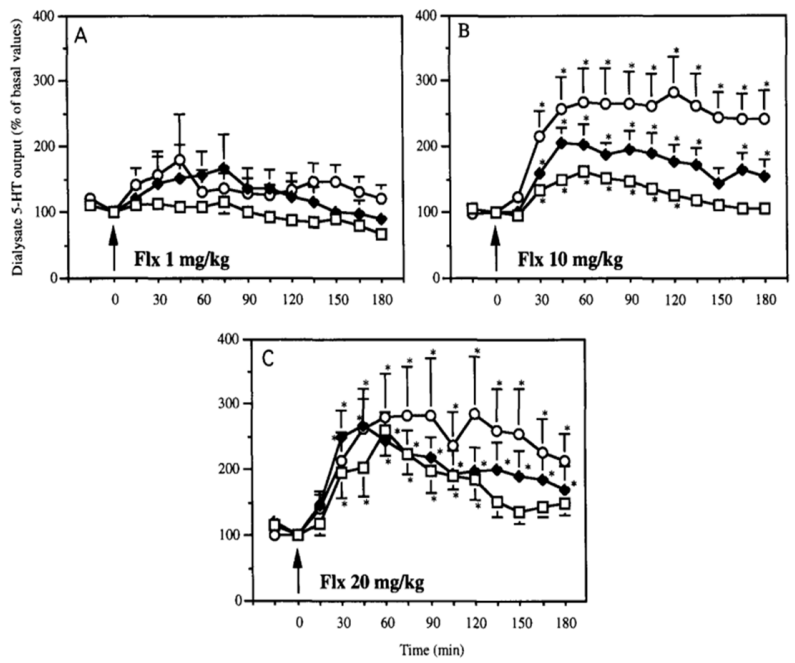
## Drug treatment and analysis

- Once baseline 5-HT levels were stable, rats received either saline control or fluoxetine at 1, 10, or 20 mg/kg.
- The dialysate samples were analyzed using HPLC with electrochemical detection (using an amperometric detector) to quantify serotonin/5-HT.
- Results were expressed as % of baseline, where pre-drug serotonin level was set as 100%.
- The total serotonin response from 0–180 minutes was summarized using AUC / Area under the curve.



# Article 2: Mechanisms of drug action at the tissues, organ or physiological level

Fig 3: Dose-dependent effect of acute fluoxetine on extracellular serotonin levels in rat brain regions

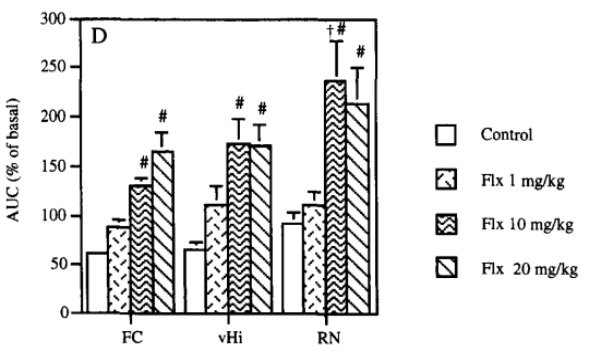


Results: Figure 3

- Extracellular serotonin/5-HT was measured in three brain regions: frontal cortex (FC), ventral hippocampus (vHi), and raphe nuclei (RN).

| Dose     | Main result                                                                                                                                                                                                                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 mg/kg  | Only a small 5-HT increase and was not statistically significant compared with control.                                                                                                                                                                                                                                |
| 10 mg/kg | A clear and statistically significant increase in 5-HT in all three brain regions. <ul style="list-style-type: none"><li>Maximum increases were: FC: 164 ± 8%, vHi: 205 ± 31%, RN: 282 ± 55% of baseline.</li></ul> The biggest effect was in the raphe nuclei, where serotonin almost tripled compared with baseline. |
| 20 mg/kg | 5-HT output increased significantly in all three brain regions. <ul style="list-style-type: none"><li>Maximum increases were: FC: 259 ± 37%, vHi: 267 ± 42%, RN: 287 ± 86% of baseline.</li></ul>                                                                                                                      |

Fig 4: AUC summary of fluoxetine-induced extracellular serotonin increases across rat brain regions



Results: Figure 4

- AUC summarized the total extracellular 5-HT response over 0–180 minutes after fluoxetine injection.
- 5-HT increase in the frontal cortex was short-lasting, from 30 to 135 minutes after drug injection. But in the ventral hippocampus and raphe nuclei, the increase lasted for the full 180-minute sampling period.

# Article 3: Mechanisms of drug action at the epidemiological level

Methods: Randomized placebo-controlled clinical trial

## Study design

- This was a 9-week acute treatment phase of a larger multiphase clinical trial.
- The study compared fluoxetine with placebo.

## Diagnostic evaluation

- Before randomisation, patients completed a 3-week diagnostic evaluation period.
- Diagnosis of MDD was confirmed using three independent diagnostic interviews.
- This helped ensure that only patients with a reliable diagnosis of nonpsychotic major depressive disorder entered the trial.

## Placebo lead-in

- After diagnosis, patients completed a 1-week single-blind placebo lead-in.
- Patients who improved during this placebo period were removed from the study.

## Participants

- Children and adolescents aged 8 to <18 years.
- All had nonpsychotic MDD.
- Patients needed a Children's Depression Rating Scale-Revised (CDRS-R) score >40 and a Clinician's Global Impressions-Severity (CGI-Severity) score  $\geq 4$ , indicating at least moderate depression.

## Treatment allocation

- After the placebo lead-in, nonresponders were randomly assigned to:
  - fluoxetine group
  - placebo group

## Fluoxetine dosing:

- Week 1: 10 mg/day fluoxetine
- Weeks 2–9: 20 mg/day fluoxetine
- The placebo group received three matching placebo capsules daily.
- Both groups took the same number of capsules.

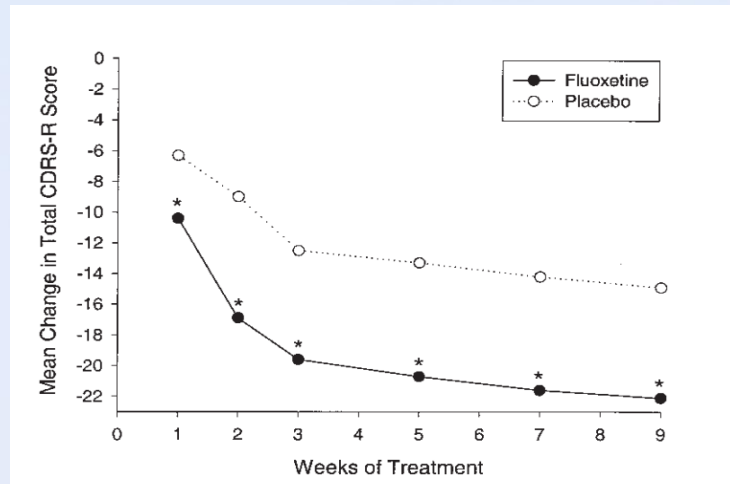
Patients returned for assessments at: weeks 1, 2, 3, 5, 7, 9

- At each visit, clinicians measured depression severity, global improvement, and adverse events.
- Main outcome measure: Lower CDRS-R score = less severe depression / improvement

Adverse events were collected in two ways:

- Nonsolicited adverse events: symptoms patients reported spontaneously
- Solicited adverse events: symptoms asked about using a side effects checklist

# Article 3: Mechanisms of drug action at the epidemiological level



| Number (and Percentage) of Patients Meeting Possible Definitions of Response |                      |                         |                      |      |
|------------------------------------------------------------------------------|----------------------|-------------------------|----------------------|------|
| Calculation Method<br>Value <sup>a</sup>                                     | Response<br>Criteria | Fluoxetine<br>(n = 109) | Placebo<br>(n = 101) | p    |
| A. Baseline – Endpoint <sup>b</sup>                                          |                      |                         |                      |      |
| Baseline                                                                     | ≥20%                 | 88 (80.7)               | 62 (61.4)            | .002 |
|                                                                              | ≥30%                 | 71 (65.1)               | 54 (53.5)            | .093 |
|                                                                              | ≥40%                 | 55 (50.5)               | 29 (28.7)            | .002 |
|                                                                              | ≥50%                 | 37 (33.9)               | 17 (16.8)            | .007 |
|                                                                              | ≥60%                 | 17 (15.6)               | 5 (5.0)              | .013 |
|                                                                              | ≥70%                 | 5 (4.6)                 | 0                    | .060 |
| B. (Baseline – 17) – (Endpoint – 17) <sup>c</sup>                            |                      |                         |                      |      |
| Baseline – 17                                                                | ≥20%                 | 95 (87.2)               | 69 (68.3)            | .001 |
|                                                                              | ≥30%                 | 86 (78.9)               | 62 (61.4)            | .006 |
|                                                                              | ≥40%                 | 77 (70.6)               | 56 (55.4)            | .031 |
|                                                                              | ≥50%                 | 63 (57.8)               | 41 (40.6)            | .014 |
|                                                                              | ≥60%                 | 56 (51.4)               | 29 (28.7)            | .001 |
|                                                                              | ≥70%                 | 40 (36.7)               | 21 (20.8)            | .015 |

<sup>a</sup> Fisher exact test.  
<sup>b</sup> Original method of calculating response rate.  
<sup>c</sup> Response rate calculation that corrects for the nonzero minimum score on the Children's Depression Rating Scale-Revised.

## Results: Table 6

- Shows how many patients achieved different levels of improvement in CDRS-R score.
- A: Original response calculation - original method used by the study.
- B: Corrected response calculation - This corrects for the fact that the CDRS-R minimum score is 17, not 0.

## A

- The original pre-defined response criterion was ≥30% decrease in CDRS-R.

At this threshold:

fluoxetine: 65.1% placebo: 53.5% p = .093  
p = .093 > .05, this difference was not statistically significant.

- Fluoxetine was significantly better than placebo at several other thresholds: ≥20%, ≥40%, ≥50%, ≥60%

## B

- Fluoxetine was significantly better than placebo at all response thresholds from ≥20% to ≥70%.

## Results : figure 5

- A decrease in CDRS-R score means the patient's depression symptoms improved.
- The fluoxetine group improved more than the placebo group throughout the study.

By week 9:

fluoxetine patients: 22-point decrease

placebo patients: 15-point decrease

- Placebo groups often improve due to clinical support, expectation, monitoring, and natural symptom variation.

**Remission result** = symptoms become low enough that the patient is considered close to no longer clinically depressed.

- In this study, remission was defined as endpoint CDRS-R score ≤28.

The results showed:

- fluoxetine remission: **41.3%**
- placebo remission: **19.8%**
- p < .01 - significantly more fluoxetine-treated patients reached remission compared with placebo.



# Discussion



## Integrated pharmacodynamic pathway

